

Testi Reflexora - 20 Luglio

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The Reflex-Policy architecture for Physical and Spatial AI.

REFLEXORA

A Physical AI architecture and IP platform — a research division of IFEVS.

We develop the architecture and intellectual property that assign each physical event to the lowest sufficient layer capable of acting safely, correctly, and energy-efficiently.

The world model predicts. The Policy Layer learns, plans, and selects. The Reflex Layer acts locally when waiting would be unsafe, inefficient, or energetically wasteful.

Discover the Architecture

Explore Technology

Placeholder

Qui verrà piazzata

l'animazione Reflex-Policy

(V10)

Placeholder · componente animato

Qui verrà integrata l'animazione interattiva

Reflex-Policy

(V10 desktop · M1 mobile)

What is

REFLEXORA?

REFLEXORA is the Physical AI research division of IFEVS dedicated to the Reflex–Policy architecture. It is being positioned for dedicated investment and is intended to evolve into an independent company as the funding and corporate transition are completed.

REFLEXORA is not simply a technology-specific device venture and not simply an AI-software venture. It is an architecture and intellectual-property platform that defines how sensing, decision, energy, communication, and actuation should cooperate in machines that operate in the physical world. The architecture is technology-neutral at the decision layer: the reflex logic can be implemented today with conventional electronics — microcontrollers, CPLDs, FPGAs, mixed-signal ASICs — and can adopt spintronic or other emerging memory technologies when their qualification justifies it.

About the company

Why

Reflex-Policy?

Advanced world models and policy models are essential for prediction, learning, planning, and complex choice. But not every physical event should wait for the richest available computation. Some events require a bounded local action before delay becomes unsafe, inefficient, or energetically wasteful.

Architecture

A Complementary

Architecture

REFLEXORA complements neuromorphic, spintronic, in-memory, analog, digital, and conventional electronic technologies. These technologies can become implementation components at different layers. REFLEXORA determines where and how they should be used across the sensing–decision–energy–actuation chain.

The architecture is technology-neutral at the decision layer. The complete reflex function — from event conditioning to deterministic rule evaluation to verified physical action — can be implemented today with established electronic components. Spintronic crossbar arrays, MRAM and other non-volatile technologies remain available as optional rule-map implementations when their system-level advantages and qualification justify adoption.

The central design rule is simple:

assign every physical event to the lowest sufficient layer that can act safely and correctly.

This reduces unnecessary ADC conversion, communication, memory access, processor wake-up and central-policy workload.

Technology

Application Areas

Robotics, Mobility & Autonomous Systems

Local impact, torque, and balance reflexes combined with predictive energy management and bounded local action for autonomous systems. The reflex logic operates with conventional electronics; spintronic rule maps are an optional future technology path.

Energy Harvesting & Storage

Local PV shadow response and module-local battery reflex balancing, isolating faults under Battery Management System supervision. The Battery Reflex Module demonstrates that the complete reflex function — measurement, rule evaluation, energy conversion, injection and verification — can be developed entirely with conventional automotive electronics.

Edge AI, IoT & Sensing

Always-on event detection, Spatial AI event-to-action contracts, and bio-inspired safety-critical edge systems. The decision layer is technology-neutral: conventional MCU, CPLD or FPGA implementations provide the direct engineering baseline, with spintronic arrays available as an optional non-volatile rule-map technology.

Applications

Intellectual

Property

REFLEXORA is supported by a coordinated series of scientific publications, manuscripts, and preprints designed to establish terminology, scientific visibility, conceptual priority, and a cross-domain research agenda for Reflex-Policy Physical AI.

Current deposited portfolio:

12 Italian patent applications, 2 EPO applications, and 1 PCT application.

Research & Publications

Corporate

Evolution

REFLEXORA is the Physical AI research division of IFEVS dedicated to the Reflex–Policy architecture. Its purpose is to define where and how intelligence should be distributed across sensing, decision, energy, communication, and actuation. The architecture is technology-neutral at the decision layer, enabling implementation with conventional electronics today while remaining open to spintronic and other emerging technologies as they reach qualification. The initiative is being positioned for dedicated funding and a planned transition into an independent company under the REFLEXORA identity and reflexora.ai domain.

REFLEXORA brings intelligence closer to the event — not by replacing advanced AI, but by ensuring that every physical action is handled by the lowest sufficient layer, with the right authority, energy, and evidence.

Company

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REFLEXORA · APPLICATIONS

Application Areas

The REFLEXORA portfolio covers robotics, mobility, autonomous systems, energy harvesting, energy storage, sensing, edge AI, and IoT. The architecture is technology-neutral at the decision layer: all application domains can be implemented today with conventional electronics, while spintronic and other emerging technologies remain available as optional enhancements. Current application studies include local PV shadow response, module-local battery reflex balancing, Spatial AI event-to-action contracts, and bio-inspired safety-critical edge systems.

Robotics and Humanoids

Local impact, torque, contact and balance reflexes that remain active while higher AI performs perception and planning. The reflex logic operates with conventional electronics — microcontrollers, CPLDs or FPGAs — and can optionally adopt spintronic rule maps when their qualification justifies it.

Automotive and Autonomous Systems

Fast current, voltage and temperature protection combined with predictive energy management and higher-level autonomy. Deterministic rule evaluation is implemented with established automotive components; spintronic arrays remain an optional technology path.

Energy Harvesting

Local photovoltaic and harvester reflexes that protect energy capture before slower optimization loops respond. The local decision logic can be realized with conventional MCU, CPLD/FPGA or ASIC implementations, with spintronic crossbars available as an optional non-volatile rule-map technology.

Energy Storage — Battery Reflex Module

Module-local battery reflex balancing: a compact mixed-signal and power-electronic assembly integrated into the Battery Module Board that identifies an eligible low cell, selects it under BMS authority, injects a bounded current, verifies the response and reports the action. The complete function can be developed entirely with conventional automotive analog, digital, mixed-signal and power-electronic components. A binary spintronic crossbar may later implement the rule map when its system-level advantages and qualification justify adoption.

Sensing

High-resolution monitoring and low-bit event conditioning for physical parameters — voltage, current, temperature, pressure, light — converted into reliable binary conditions by conventional comparators, AFEs and threshold circuits. Event detection does not require exotic device technologies.

Edge AI and IoT

Always-on event detection, selective wake-up and compressed communication for energy-constrained devices. The decision layer is technology-neutral: conventional MCU or FPGA implementations provide the direct engineering baseline, with spintronic arrays and other non-volatile memories available as optional future paths.

[Read the research](#)

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REFLEXORA · ARCHITECTURE

The Reflex–Policy Architecture

Advanced world models and policy models are essential for prediction, learning, planning, and complex choice. But not every physical event should wait for the richest available computation. Some events require a bounded local action before delay becomes unsafe, inefficient, or energetically wasteful.

Reflex–Policy separates the urgent local path from slower supervisory intelligence. The Policy Layer defines permissions, context, and rules. The Reflex Layer executes only the first permitted action, verifies the physical response, and reports a compact event-action trace. The architecture is technology-neutral at the decision layer: reflex logic can be implemented with conventional electronics, and can adopt spintronic or other emerging technologies as an optional enhancement.

What makes the architecture distinctive

Event-to-action partitioning

Not every sensor event should travel through the richest available model. The architecture identifies which events require immediate local action, which require policy interpretation, and which require world-model reasoning.

Bounded authority

A reflex does not receive unlimited control. It acts inside a permission envelope defined by safety, operating state, configuration, and supervisory policy.

Measured closure

The architecture includes the actuator command and the physical feedback that confirms whether the commanded response actually occurred.

Compact observability

Local actions produce event-action traces: what happened, what was permitted, what was commanded, what was measured, and whether containment or fallback was required.

Energy proportionality

The EROIE framing asks whether the energy protected, recovered, or made useful by an action justifies the sensing, inference, electronics, and actuation energy spent to obtain it.

World model as teacher

High-cost models may learn or optimize rules offline or periodically, then transfer bounded rule maps to local execution layers that do not require full-model latency at runtime.

Technology-neutral decision layer

The reflex logic can be implemented with conventional electronic components — microcontrollers, CPLDs, FPGAs, mixed-signal ASICs — or with optional spintronic crossbar arrays and other non-volatile memories. The architecture does not depend on any single device technology.

Explore the technology

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REFLEXORA · COMPANY

About REFLEXORA

REFLEXORA is the Physical AI research division of IFEVS dedicated to the Reflex-Policy architecture. It is being positioned for dedicated investment and is intended to evolve into an independent company as the funding and corporate transition are completed.

REFLEXORA is not simply a technology-specific device venture and not simply an AI-software venture. It is an architecture and intellectual-property platform that defines how sensing, decision, energy, communication, and actuation should cooperate in machines that operate in the physical world. The architecture is technology-neutral at the decision layer: reflex logic can be implemented with conventional electronics today, and can adopt spintronic or other emerging technologies as an optional enhancement.

REFLEXORA is the Physical AI research division of IFEVS dedicated to the Reflex-Policy architecture. Its purpose is to define where and how intelligence should be distributed across sensing, decision, energy, communication, and actuation, so that urgent physical events can receive bounded local action while policy and world models remain responsible for learning, planning, prediction, and rule updates. The architecture is

technology-neutral at the decision layer: the complete reflex function can be implemented today with conventional electronic components, and spintronic or other emerging technologies can be adopted as optional enhancements when their qualification justifies it. REFLEXORA is supported by 12 deposited Italian patent applications, two deposited EPO applications, one deposited PCT application, and a coordinated scientific publication and preprint program. The initiative is being positioned for dedicated funding and a planned transition into an independent company under the REFLEXORA identity and reflexora.ai domain.

REFLEXORA is a research division of

IFEVS

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Get in touch

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REFLEXORA · CONTACT

Get in

Touch

REFLEXORA is being positioned for dedicated investment and partnership. For collaboration, investment, or research inquiries, reach us directly by email.

contact@reflexora.ai

REFLEXORA is a research division of

IFEVS

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REFLEXORA · RESEARCH

Research & Scientific Foundation

REFLEXORA is supported by a coordinated series of scientific publications, manuscripts, and preprints designed to establish terminology, scientific visibility, conceptual priority, and a cross-domain research agenda for Reflex–Policy Physical AI.

Perspective papers

REFLEXORA's architecture is documented in two Perspective papers covering:

the nature-inspired five-layer Reflex–Policy framework, mutual protection and EROIE; latency, throughput, bandwidth, world models and the role of binary and multilevel spintronic devices.

The papers are being made publicly available through [Preprints.org](https://preprints.org) to establish an open scientific foundation for the Reflex–Policy field.

Quantitative architecture

Illustrative application studies indicate that local event compression can reduce urgent upstream communication by one to three orders of magnitude, depending on channel count, sampling rate and event frequency. These figures are design examples and must be validated for each application.

Technology-neutral implementation

The technical report on the Battery Reflex-Balance Module describes how the complete reflex function — from measurement and event conditioning through deterministic rule evaluation to verified physical action — can be implemented today with conventional automotive electronics. The same approach applies across all application domains covered by the Reflex–Policy architecture: energy storage, energy harvesting, robotics, edge AI and IoT. Spintronic crossbar arrays and other emerging non-volatile technologies remain available as optional rule-map implementations, to be adopted when their system-level advantages and qualification justify it.

About the company

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REFLEXORA · TECHNOLOGY

A Technology-Neutral Architecture

REFLEXORA complements neuromorphic, spintronic, in-memory, analog, digital, and conventional electronic technologies. These technologies can become implementation components at different layers. REFLEXORA determines where and how they should be used across the sensing–decision–energy–actuation chain.

REFLEXORA is not another accelerator. It is the architecture that decides which events need acceleration, which events need only a low-bit local rule, and how the resulting physical action remains safe, observable, and energy-proportional. The architecture is technology-neutral at the decision layer.

Conventional Electronic Implementation

The complete reflex function can be implemented today with established components. The conventional implementation path demonstrates that the Reflex–Policy architecture does not require exotic device technologies to deliver its core value.

The local decision logic can be realized in several mature ways:

A microcontroller reads sensor flags, evaluates a small deterministic rule table and commands the output selector.

A CPLD or FPGA implements the same rules in parallel with explicit one-hot logic and hardware timing.

For production, comparator banks, rule logic, timers, priority encoders, interlocks and communication interfaces can be integrated into a mixed-signal ASIC or next-generation controller.

Non-volatile configuration may be stored in automotive flash, EEPROM, embedded MRAM or another qualified memory.

This conventional path provides the direct engineering baseline across all application domains — battery systems, energy harvesting, robotics, edge AI and IoT.

Technology-Neutral Rule-Map Alternatives

The decision and rule-map block can be implemented with multiple technologies, selected according to the requirements of each application:

Conventional approaches

MCU with a verified rule table and hardware output interlocks

CPLD or FPGA with parallel logic, deterministic timing and one-hot arbitration

Mixed-signal ASIC or configurable CMOS logic with embedded flash, EEPROM or another qualified non-volatile memory

Hardwired comparator and state-machine logic for a fixed-function, minimum-complexity implementation

Optional emerging technologies

Embedded MRAM or binary MTJ crossbar

RRAM, FeFET, phase-change memory

Other non-volatile low-resolution arrays

These can be adopted when their system-level advantages and qualification justify it.

Spintronic Devices as an Optional Technology Path

Binary and multilevel spintronic devices serve different roles within the architecture, but their adoption is optional and can be assessed independently for each application.

Binary spintronic devices

A binary spintronic crossbar may be used as an alternative implementation of the decision and rule-map block. It is not required for the Reflex architecture. Binary spintronic devices are suited to fast and inspectable decisions such as:

enable or inhibit

connect or bypass

wake or sleep

safe or unsafe

permit or block

normal or fault

The binary spintronic array provides only the stored low-bit mapping between local conditions and candidate actions. It does not regulate or carry power. Its practical use should be assessed based on foundry availability, automotive temperature range, endurance, retention, sense margins, write energy, qualification and long-term supply.

Multilevel spintronic devices

Multilevel spintronic devices support richer weighted operations, including:

sensor fusion

neural inference

classification

optimization

model approximation

policy acceleration

REFLEXORA studies how these device classes can cooperate within the same architecture, rather than treating all forms of spintronic computation as interchangeable.

Until system-level advantages and qualification are demonstrated for the intended application, the conventional electronic implementation remains the direct engineering baseline.

Decision–Power Path Isolation

The reflex module remains in the decision path. The principal energy remains in the power path. A small, low-energy rule signal controls a much larger physical action — like a pilot valve or railway switch — allowing reflex intelligence to be integrated with conventional and certifiable power electronics. This separation is independent of the decision-layer technology: whether the rule map is implemented with a microcontroller, an FPGA, a mixed-signal ASIC or an optional spintronic crossbar, the physical energy-transfer chain remains a conventional protected power-electronic path.

World Models as Teachers

World models explore thousands of situations during simulation, training or periodic optimization. Their conclusions are converted into bounded binary rules, safe thresholds, permission maps and multilevel weight matrices. The world model does not need to remain in every urgent control loop.

See the applications